

# STUDENT MANAGEMENT SYSTEM

Java Internship Project

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Role: Java Developer Intern

Organization: Codec Technologies

# Internship Details

- Organization: Codec Technologies
- Internship Role: Java Developer Intern
- Internship Type: Project-Based Internship
- Project Title: Student Management System using Java
- Internship Period: July 2026

# Project Introduction

- A console-based Java application for managing student records.
- Designed to perform common student information operations efficiently.
- Uses Object-Oriented Programming principles.
- Supports persistent storage using file handling.
- Provides a simple menu-driven interface.

# Problem Statement

- Managing student records manually can be repetitive and difficult.
- Users need an easy way to add, search, update and delete records.
- The project provides a structured digital solution for basic student record management.

# Project Objectives

- Develop a practical application using Core Java.
- Implement CRUD operations on student records.
- Apply Object-Oriented Programming concepts.
- Use Java Collections Framework for record management.
- Implement input validation and error handling.
- Store student data using file handling.

# Technologies Used

- Java Programming Language
- Object-Oriented Programming (OOP)
- ArrayList / Java Collections Framework
- File Handling and Serialization
- Scanner for User Input
- Exception Handling

# System Architecture



# Main Features

- Add Student
- View All Students
- Search Student by ID
- Update Student Details
- Delete Student Record
- Duplicate ID Validation
- Input Validation
- Persistent Data Storage

# OOP Concepts Used

- Classes and Objects
- Encapsulation using private fields and getter/setter methods
- Constructors for object initialization
- Methods for modular functionality
- Separation of data model, business logic and user interface

# Project Structure

- Student-Management-System/
  - └─ src/
    - | └─ Main.java
    - | └─ Student.java
    - | └─ StudentManager.java
  - └─ README.md
  - └─ .gitignore
  - └─ students.dat (created automatically)

# Working of the System

- User selects an option from the main menu.
- Main.java receives and validates user input.
- StudentManager processes the requested operation.
- Student objects represent individual student records.
- Changes are saved to a local file for future use.

# Testing and Results

- Student addition tested successfully.
- Duplicate student IDs are rejected.
- Search operation returns matching student records.
- Update operation modifies existing records.
- Delete operation removes records.
- Saved data can be loaded when the application restarts.

# Future Scope

- Add a graphical user interface using JavaFX or Swing.
- Integrate MySQL database using JDBC.
- Implement user authentication.
- Add sorting, filtering and reporting.
- Develop a web-based version.
- Add role-based access for administrators and staff.

# Conclusion

- The Student Management System successfully demonstrates practical Java programming.
- The project implements CRUD operations and file-based persistence.
- It applies important concepts such as OOP, collections, validation and exception handling.
- The project provides a strong foundation for developing larger Java applications.

**THANK YOU**

Questions & Discussion